Project Proposal: Creating Tailored Honeypots with OpenAI's, Shodan.io and LangChain

1. Introduction

The purpose of this project is to develop a framework for creating honeypots and deception technology using OpenAI's model and LangChain. The objective is to misdirect and trap attackers by deploying customized and lightweight honeypots based on provided public IP ranges for customers or clients.

1a. Problem Statement: Assessing Vulnerability and Network Scans

Can we utilize OpenAI's Language Model (LLM) to analyze vulnerability data scans and network scans, such as port scans? The project aims to leverage LLM's capabilities to assess and process the collected scan data effectively.

1b. Problem Statement: Lightweight Deployable Honeypots

Can OpenAI's GPT4 model generate lightweight and tailored honeypots based on port scans? The project seeks to explore the potential of using GPT4 to create deployable honeypots that can effectively deceive attackers and gather valuable information.

1c. Problem Statement: Ease of Use and Integration

Can we make the project easily accessible and usable for anyone? This includes providing integrations or plugins to OpenAI's existing framework, simplifying the deployment process, and offering user-friendly interfaces.

2. Methodologies and Approaches

2a. Data Collection and Analysis

The project will utilize existing network scans obtained from Shodan.io for external web domains and IP address ranges. These scan data will be fed into OpenAI's language

model to summarize and identify optimal locations for integrating honeypots.

2b. Lightweight Honeypot Generation

Customized lightweight honeypots will be generated using scripting languages such as PowerShell, Python, Perl, and Bash. The generated scripts will be tailored to the identified honeypot locations and designed for easy deployment.

3. Expected Results

The project aims to provide the following outcomes:

1. Web domain or IP range input functionality with LangChain Agent: Users can feed a web domain or IP range, which will be crawled using a custom LangChain Agent connected to OpenAI's GPT4 model.
2. Interactive Data Analysis: Users can interact with the collected data using the LLM-like chatGPT interface. This allows for intuitive exploration and analysis of the data to identify potential honeypot locations.
3. Lightweight Honeypot Generation: Users can select the optimal honeypot locations suggested by the system. The project will generate easily deployable scripts along with instructions on running them in a dockerized container.
4. User Guidance and Best Practices: The project will provide information and guidelines to ensure users can effectively utilize the honeypots. This includes instructions on proper configuration, monitoring, and maintenance.
5. Data Import and Threat Profiling: Data collected from the honeypots can be seamlessly imported into OpenAI's platform to generate summaries and build potential threat profiles. This enhances the overall understanding of attacker behavior and enables proactive threat mitigation.

In conclusion, this project aims to develop a comprehensive framework for creating and deploying honeypots using OpenAI's model and LangChain. By leveraging the power of AI and deception technology, the project seeks to enhance cybersecurity measures and provide valuable insights for effective threat mitigation.